

① Source	Geothermal	Petroleum
Source	Heat sources earth/magma	Hydrocarbon ↳ Buried organics
Reservoir	Fractured rock, Fault	Porous Sediment Rock
Cap	Clay cap/ Clay layers Smectite Clay	Impermeable Shale
Fluid	Steam + Brine	Oil & Gas
Pressure	High hydrostatic + high temp. Gradient	Moderate to high

- ② Well control Sys. : menjaga aliran ^{fluida} tak terkendali dari well bore
Contoh : BOP
- Rotating System : Sistem pergerakan berputar untuk memecah/membek batuan
con : top drive
- Hoisting System : untuk mengangkat drill pipe & drill string
con : drawworks



SHERATON

EST. 1937

② Circulation System : untuk mengontrolasi mud & fluid
pencampuran

contoh : mud pump

Power System : untuk menghasilkan power / listrik
pada rig

con : PLTS + Baterai

③

a. MW = 10,9 ppg TD : 9,015 MD
grad. = 0,578 PSI/ft TVD : 8,505 TMD

- FP pressure = $0,578 \text{ PSI/ft} \times 8,505 \text{ ft} = 4,916,9 \text{ PSI}$

- H.P. = $0,052 \times 10,9 \text{ ppg} \times 8,505 \text{ ft} = 4,856,8 \text{ PSI}$

$\Delta P = +60,1 \text{ PSI}$ (FP > HP) = underbalanced cond.

b. FP = $4,916,9 \text{ PSI} = 0,052 \times \text{MW}_{\text{min.}} \times 8,505$

$\text{MW}_{\text{min.}} = \frac{4916,9}{442,26} = 11,12 \text{ ppg}$ (minimum MW
untuk menjaga
underbalance)

④ a. casing shoe D = 9000 MD / 0500 TVD

Toc Depth = 2000 ft diatas 9-5/8" shoe

= $6500 - 2000 = 4,500 \text{ MD} / 3500 \text{ TVD}$

Cement h. = $8500 - 3500 = 5,000 \text{ ft}$



SHERATON

EST 1937

* Sec. 1 // 7" shoe = 8500 TVD Float collar = 8470 TVD

$$H_2 = 8500 - 8470 = 30 \text{ ft}$$

Annular capacity = (7" casing OD vs 8.6" hole):

$$\begin{aligned} & (D_h^2 - D_p^2) / 1029.4 = (8.6^2 - 7.0^2) / 1029.4 \\ & = 0.024 \text{ bbl/ft} \end{aligned}$$

$$\text{Vol} = 0.024 \times 30 = 0.72 \text{ bbl}$$

* Sec. 2 // Float collar (8470 TVD) TO TOC (3500 TVD)

$$h = 8470 - 3500$$

annular capacity = 0.024 bbl/ft

$$\text{Vol.} = 0.024 \times 4.970 = 120.6 \text{ bbl}$$

$$\text{total} = 0.72 + 120.6 = \underline{\underline{121 \text{ bbl}}}$$

b. spacer

$$= (10^2) / 1029.4$$

$$= (8.95^2) / 1029.4 = 0.076 \text{ bbl/ft}$$

- Sec. 1

$$h = 5000 \text{ ft}$$

$$V = 5000 \times 0.076 = 380.15 \text{ bbl}$$

- Sec 2

$$h = 5000 - 3500 = 1500 \text{ ft}$$

$$\text{spacer} = (6.185^2) / 1029.4 = 0.037 \text{ bbl/ft}$$

$$V = 1500 \times 0.037$$

$$= 55.74 \text{ bbl}$$

SHERATON

EST. 1937

$$\text{total} = 380.15 + 55.74 = \underline{\underline{436 \text{ bbl}}}$$

C. 7" casing cap = 0.037 bbl/ft

TOP = 0 TVD

7" casing shoe h = 8500 TVD

Displacement h = 8500 TVD

$V = 0.037 \times 8500 = 316 \text{ bbl}$

⑤ J-type KOP = 430 TVD

$\theta_{\text{max}} = 62^\circ$ DLS = $2.5^\circ/1000 \text{ ft}$

Int. turn section = $30^\circ \text{ Inc. } / 720 \text{ ft MD}$

$\text{DLS} = \frac{5729.58^\circ}{R}$

$R = \frac{5729.58}{2.5} = 2291.83 \text{ ft}$

$\Delta\theta_1 = 30^\circ - 0^\circ = 30^\circ \text{ (KOP)}$

$\Delta\theta_2 = 62^\circ - 30^\circ = 32^\circ \text{ (max. inclination)}$

$\text{MD} = \frac{R \times \Delta\theta}{57.296}$

$\text{MD}_1 = \frac{2291.83 \times 30}{57.296} = 1200.83 \text{ ft}$

$\text{MD}_3 = \frac{2291.83 \times 32}{57.296} = 1280.88 \text{ ft}$



$\text{MD}_2 = 720 \text{ ft (like Hawaii)}$

SHERATON
EST. 1937

total = KOP + MD₁₂₃

= 430 + 1201 + 720

+ 1280 = 3631 ft